



Bhabha University, Bhopal (M.P.)

Scheme of Examination

Program: Bachelor of Technology
Semester : IV

Branch: Information Technology
Specialization: Information Technology

Session: 2020-21

S. No .	Subject Code	Subject Name	Maximum Marks Allotted					Total Marks	Contact Hours per week			Total Credits
			Theory			Practical						
			End Sem.	Mid Sem. Exam.	Quiz/ Assignment	End Sem.	Term work					
							Lab Work & Sessional					
1.	IT-41	Discrete Structure	70	20	10	-	-	100	3	1	-	4
2.	IT-42	Computer Architecture	70	20	10	30	20	150	2	1	2	4
3.	IT-43	Analysis and Design of Algorithm	70	20	10	30	20	150	3	1	2	5
4.	IT-44	Analog & Digital Communication	70	20	10	30	20	150	3	1	2	5
5.	IT-45	DatabaseManagementSystem	70	20	10	30	20	150	3	0	2	4
6.	IT-46	Programming PracticesJAVA	-	-	-	30	20	50	-	-	4	2
		Total	350	100	50	150	100	750	14	4	12	24

* End Semester Examination will be of three hours End Semester Examination will be of three hours.

PRACTICAL SUBJECT IT- 42 : Computer Organization and architecture
IT-44:-Analog and Digital communication
IT-46 :Programming Practices JAVA

IT-43:-ADA
IT-45:-DBMS

1 Hr Lecture	1 Hr Tutorial	2 Hr Practical
1 Credit	1 Credit	1 Credit



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**COURSE CURRICULUM
FACULTY OF ENGINEERING**

Undergraduate Programme

(Courses effective from Academic Year 2020-21)

**SYLLABUS OF COURSES TO BE OFFERED
“Bachelor of Technology”**

Disclaimer: The undergraduate programme in Bachelor of Technology syllabus & scheme is uploaded as given by the Faculty concerned to the Academic Council. The same has been approved as it is by the Academic Council on 02.07.2018.

Dean faculty of Engineering, Bhabha University

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IT-41 Discrete Structure

UNIT I:- Set Theory, Relation, Function, Theorem Proving Techniques : Set Theory: Definition of sets, countable and uncountable sets, Venn Diagrams, proofs of some general identities on sets Relation: Definition, types of relation, composition of relations, Pictorial representation of relation, Equivalence relation, Partial ordering relation, Job-Scheduling problem Function: Definition, type of functions, one to one, into and onto function, inverse function, composition of functions, recursively defined functions, pigeonhole principle. Theorem proving Techniques: Mathematical induction, Proof by contradiction.

UNIT II:- Algebraic Structures: Definition, Properties, types: Semi Groups, Monoid, Groups, Abelian group, properties of groups, Subgroup, cyclic groups, Co-sets, factor group, Permutation groups, Normal subgroup, Homomorphism and isomorphism of Groups, example and standard results, Rings and Fields: definition and standard results.

UNIT III:- Propositional Logic: Proposition, First order logic, Basic logical operation, truth tables, tautologies, Contradictions, Algebra of Proposition, logical implications, logical equivalence, predicates, Normal Forms, Universal and existential quantifiers. Introduction to finite state machine Finite state machines as models of physical system equivalence machines, Finite state machines as language recognizers

UNIT IV:- Graph Theory: Introduction and basic terminology of graphs, Planer graphs, Multigraphs and weighted graphs, Isomorphic graphs, Paths, Cycles and connectivity, Shortest path in weighted graph, Introduction to Eulerian paths and circuits, Hamiltonian paths and circuits, Graph coloring, chromatic number, Isomorphism and Homomorphism of graphs.

UNIT V:- Posets, Hasse Diagram and Lattices: Introduction, ordered set, Hasse diagram of partially ordered set, isomorphic ordered set, well ordered set, properties of Lattices, bounded and complemented lattices.

Combinatorics: Introduction, Permutation and combination, Binomial Theorem, Multinomial Coefficients Recurrence Relation and Generating Function: Introduction to Recurrence Relation and Recursive algorithms, Linear recurrence relations with constant coefficients, Homogeneous solutions, Particular solutions, Total solutions, Generating functions, Solution by method of generating functions.



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REFERENCES

- 1) C.L.Liu, "Elements of Discrete Mathematics" Tata Mc Graw-Hill Edition.
- 2) Trembley, J.P & Manohar; "Discrete Mathematical Structure with Application CS", McGrawHill.
- 3) Kenneth H. Rosen, "Discrete Mathematics and its applications", McGrawHill.
- 4) Bisht, "Discrete Mathematics", Oxford University Press
- 5) Biswal, "Discrete Mathematics & Graph Theory", PHI





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IT-42 Computer Architecture

Unit I:-Computer architecture and organization, computer generations, von Neumann model, CPU organization, CPU organization, register organization, Various CPU register, Register Transfer, Bus and Memory Transfers, Arithmetic, Logic and Shift micro-operations, Arithmetic logic shift unit.

Unit II:-The arithmetic and logic unit, Fixed-Point representation: integer representation, sign-magnitude, 1's and 2's complement and range, Integer arithmetic: negation, addition and subtraction, multiplication, division, Floating-Point representation, Floating-Point arithmetic, Hardwired micro-programmed control unit, Control memory, Micro-program sequence.

Unit III:-Central Processing Unit (CPU), Stack Organization, Memory Stack, Reverse Polish Notation. Instruction Formats, Zero, One, Two, Three- Address Instructions, RISC Instructions and CISC Characteristics, Addressing Modes, Modes of Transfer, Priority Interrupt, Daisy Chaining, DMA, Input-Output Processor (IOP).

Unit IV:-Computer memory system, Memory hierarchy, main memory: RAM, ROM chip, auxiliary and associative memory, Cache memory: associative mapping, direct mapping, set-associative mapping, write policy, cache performance, Virtual memory: address space, memory space, address mapping, paging and segmentation, TLB, page fault, effective access time, replacement algorithm.

Unit V:-Parallel Processing, Pipelining General Consideration, Arithmetic Pipeline, and Instruction Pipeline, Vector Operations, Matrix Multiplication, and Memory Interleaving, Multiprocessors, Characteristics of Multiprocessors.

LIST OF EXPERIMENTS

- 1) Study of Multiplexer and Demultiplexer
- 2) Study of Half Adder and Subtractor
- 3) Study of Full Adder and Subtractor
- 4) WAP to add two 8 bit numbers and store the result at memory location 2000
- 5) WAP to multiply two 8 bit numbers stored at memory location 2000 and 2001 and stores the result at memory location 2000 and 2001.
- 6) WAP to add two 16-bit numbers. Store the result at memory address starting from 2000.
- 7) WAP which tests if any bit is '0' in a data byte specified at an address 2000. If it is so, 00 would be stored at address 2001 and if not so then FF should be stored at the same address.



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REFERENCES

- 1) M. Morris Mano, "Computer System Architecture", Pearson.
- 2) Dr.M.Usha,T.S.Srikanth, "ComputerSystemArchitectureandOrganization", Wiley India.
- 3) William Stallings, "Computer Organization and Architecture", Pearson.
- 4) V. Rajaraman, T. Radhakrishnan, "Computer Organization and Architecture", PHI.





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IT-43 Analysis and Design of Algorithm

UnitI:-Algorithms, Designing algorithms, analyzing algorithms, asymptotic notations, heap and heap sort. Introduction to divide and conquer technique, analysis, design and comparison of various algorithms based on this technique, example binary search, merge sort, quick sort, strassen's matrix multiplication.

UnitII:-Study of Greedy strategy, examples of greedy method like optimal merge patterns, Huffman coding, minimum spanning trees, knapsack problem, job sequencing with deadlines, single source shortest path algorithm, etc.

UnitIII:-Concept of dynamic programming, problems based on this approach such as 0/1 knapsack, multistage graph, reliability design, Floyd-Warshall algorithm, etc.

UnitIV:-Backtracking concept and its examples like 8 queen's problem, Hamiltonian cycle, Graph coloring problem etc. Introduction to branch & bound method, examples of branch and bound method like traveling salesman problem etc. Meaning of lower bound theory and its use in solving algebraic problem, introduction to parallel algorithms.

UnitV:-Binary search trees, height balanced trees, 2-3 trees, B-trees, basic search and traversal techniques for trees and graphs (In order, preorder, postorder, DFS, BFS), NP-completeness.

LIST OF EXPERIMENTS

- 1) Write a program for Iterative and Recursive Binary Search.
- 2) Write a program for Merge Sort.
- 3) Write a program for Quick Sort.
- 4) Write a program for Strassen's Matrix Multiplication.
- 5) Write a program for optimal merge patterns.
- 6) Write a program for Huffman coding.
- 7) Write a program for minimum spanning trees using Kruskal's algorithm.
- 8) Write a program for minimum spanning trees using Prim's algorithm.
- 9) Write a program for single sources shortest path algorithm.
- 10) Write a program for Floyd-Warshall algorithm.
- 11) Write a program for traveling salesman problem.
- 12) Write a program for Hamiltonian cycle problem.

REFERENCES

1. Cormen Thomas, Leiserson CE, Rivest RL; Introduction to Algorithms; PHI.
2. Horowitz & Sahani; Analysis & Design of Algorithm
3. Dasgupta; algorithms; TMH
4. Ullmann; Analysis & Design of Algorithm;
5. Michael T Goodrich, Roberto Tamassia, Algorithm Design, Wiley India



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IT-44 Analog & Digital Communication

Unit I:- Signals and Systems Block diagram of a communication system, signal-definition, types of signals continuous, discrete, deterministic, non-deterministic, periodic, non-periodic, energy, power, analog and digital signals. Electromagnetic Spectra, Standard signals- DC, sinusoidal, unit step, ramp, signum, rectangular pulse, impulse(delta) signal. System definition, classification of systems, linear, nonlinear, time variant, time invariant, causal, non causal, stable and unstable systems. Fourier transforms: Time domain and frequency domain representation of signal, Fourier Transform and its properties, conditions for existence, Transform of Gate, unit step, constant, impulse, sine and cosine wave. Shifting property of delta function, convolution, time and frequency convolution theorems.

Unit II:- Amplitude modulation Modulation, need of modulation, types of modulation techniques, amplitude modulation (DSB-FC), modulation index, frequency spectrum of AM wave, linear and over modulation, power relation in AM, transmission efficiency, modulation by a complex signal, bandwidth of AM, AM modulators, square law and switching modulator, advantages and disadvantages of AM. Demodulation of AM: Suppressed carrier amplitude modulation systems, DSB-SC, SSB-SC, VSB-SC systems, comparison of various amplitude modulation systems. Demodulation of AM, square law and envelope detector, synchronous detection of AM, Low and high power AM transmitters, AM receivers, TRF and super heterodyne receivers, sensitivity, selectivity and fidelity of receivers.

Unit III:- Angle modulation Introduction and types of angle modulation, frequency modulation, frequency deviation, modulation index, deviation ratio, bandwidth requirement of FM wave, types of FM. Phase modulation, difference between FM and PM, Direct and indirect method of FM generation, FM demodulators- slope detector, Foster seeley discriminator, ratio detector. Introduction to pulse modulation systems.

Unit IV:- Sampling of signal, sampling theorem for low pass and Band pass signal, Pulse amplitude modulation (PAM), Time division, multiplexing (TDM). Channel Bandwidth for PAM-TDM signal. Types of sampling instantaneous, Natural and flat top, Aperture effect, Introduction to pulse position and pulse duration modulations, Digital signal, Quantization, Quantization error, Pulse code modulation, signal to noise ratio, Companding, Data rate and Baud rate, Bit rate, multiplexed PCM signal, Differential PCM (DPCM), Delta Modulation (DM) and Adaptive Delta Modulation (ADM), comparison of various systems.

Unit V:- Digital modulations techniques, Generation, detection, equation and Bandwidth of amplitude shift keying (ASK) Binary Phase Shift keying (BPSK), Differential phase shift keying (DPSK), offset and non offset quadrature phase shift keying (QPSK), M-Ary PSK, Binary frequency Shift Keying (BFSK), M-Ary FSK Quadrature Amplitude modulation (QAM).



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LIST OF EXPERIMENTS

- 1) AM Modulation and Demodulation (EnvelopeDetector)
- 2) Frequency modulation using reactancemodulator.
- 3) Frequency modulation using varactormodulator.
- 4) Pulse Amplitude Modulation andDemodulation
- 5) Pre-emphasis andDe-emphasis
- 6) AnalogMultiplexing.
- 7) Amplitude Modulation usingPspice
- 8) Receiver characteristics (selectivity, sensitivity,fidelity).
- 9) Operation of foster-seeley loopdetector.
- 10) Operation of ratiodelector.

REFERENCES

- 1) Singh &Sapre, “Communication Systems”,TMH.
- 2) Taub Schilling, “Principles of Communication Systems”,TMH.
- 3) W. Tomasi “Electronic Communications Systems”, Pearson Education Pvt.Ltd.
- 4) Taub & shilling, “Communication Systems”,TMH.
- 5) Abhay Gandhi, “Analog and Digital Communication”, CENGAGELearning.





IT-45 Data Base Management System

UNIT I:- Basic Concepts Introduction to DBMS, File system vs DBMS, Advantages of database systems, Database System architecture, Data models, Schemas and instances, Data independence, Functions of DBA and designer, Entities and attributes, Entity types, Key attributes, Relationships, Defining the E-R diagram of database.

UNIT II:- Relational Model Structure of relational databases, Domains, Relations, Relational algebra – fundamental operators and syntax, relational algebra queries, Entity-Relationship model: Basic concepts, Design process, constraints, Keys, Design issues, E-R diagrams, weak entity sets, extended E-R features – generalization, specialization and aggregation

UNIT III:- SQL Data definition in SQL, update statements and views in SQL: Data storage and definitions, Data retrieval queries and update statements, Query Processing & Query Optimization: Overview, measures of query cost, selection operation, sorting, join, evaluation of expressions, transformation of relational expressions, estimating statistics of expression results, evaluation plans. Case Study of ORACLE and DB2.

UNIT IV:- Relational Database design Functional Dependency – definition, trivial and non-trivial FD, closure of FD set, closure of attributes, irreducible set of FD, Normalization – 1NF, 2NF, 3NF, Decomposition using FD- dependency preservation, lossless join, BCNF, Multi-valued dependency, 4NF, Join dependency and 5NF

UNIT V:- Introduction of transaction, transaction processing and recovery, Concurrency control: Lock management, specialized locking techniques, concurrency control without locking, Protection and Security Introduction to: Distributed databases, Basic concepts of object oriented data base system.

LIST OF EXPERIMENTS

- 1) To perform various SQL Commands of DDL, DML, DCL.
- 2) Write SQL Commands such as Insertion, deletion and updation for any schema.
- 3) To execute Nested Queries, Join Queries, order-by, having clause and string operation.
- 4) To perform set operators like Union, Intersect, Minus on a set of tables.
- 5) To execute various commands for GROUP functions (avg, count, max, min, Sum).
- 6) Write a PL/SQL block for transaction application using Triggers.
- 7) Write a DBMS program to prepare report for an application using function.
- 8) Designing of various Input screens/Forms.
- 9) Create reports using database connectivity of Front end with backend.
- 10) Create database Design with normalization and implementing in any application.



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REFERENCES

- 1) Korth, Silbertz, Sudarshan, “Database Concepts”, McGraw Hill.
- 2) Elmasri, Navathe, “Fundamentals of Database Systems”, Pearson.
- 3) Ivan Bayross, “SQL, PL/SQL the Programming Language of Oracle”, BPB publications.
- 4) S. Sharma, J. Agrawal, S. Agrawal, “Advanced Database Management System”, Dreamtech Press.
- 5) Leon & Leon, “Fundamental of Data Base Management System”, TMH





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IT-46 Programming Practices JAVA

UNIT I:-Basic Java Features - C++ Vs JAVA, JAVA virtual machine, Constant & Variables, Data Types, Class, Methods, Objects, Strings and Arrays, Type Casting, Operators, Precedence relations, Control Statements, Exception Handling, File and Streams, Visibility, Constructors, Operator and Methods Overloading, Static Members, Inheritance: Polymorphism, Abstract methods and Classes

UNIT II:-JavaCollectiveFrameWork-DataStructures:Introduction,Type-WrapperClassesfor Primitive Types, Dynamic Memory Allocation, Linked List, Stack, Queues, Trees, Generics: Introduction, Overloading Generic Methods, Generic Classes, Collections: Interface Collection and Class Collections, Lists, Array List and Iterator, Linked List, Vector. Collections Algorithms: Algorithm sorts, Algorithm shuffle, Algorithms reverse, fill, copy, max and min Algorithm binary Search, Algorithms add All, Stack Class of Package java. Util, Class Priority Queue and Interface Queue, Maps, Properties Class, Un- modifiableCollections.

UNIT III:-Advance Java Features - Multithreading: Thread States, Priorities and Thread Scheduling, Life Cycle of a Thread, Thread Synchronization, Creating and Executing Threads, Multithreading with GUI,MonitorsandMonitor Locks. Networking: Manipulating URLs, Reading a file on a Web Server, Socket programming, Security and the Network, RMI, Networking, Accessing Databases with JDBC: Relational Database, SQL, MySQL, Oracle

UNIT IV:-Advance Java Technologies - Servlets: Overview and Architecture, Setting Up the Apache Tomcat Server, Handling HTTP get Requests, Deploying a web Application, Multitier Applications, Using JDBC from a Servlet, Java Server Pages (JSP): Overview, First JSPEXample, Implicit Objects, Scripting, Standard Actions, Directives, Multimedia: Applets and Application: Loading, Displaying and Scaling Images, Animating a Series of Images, Loading and playing Audio clips

UNIT V:-Advance Web/Internet Programming (Overview): J2ME, J2EE, EJB, XML.





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LIST OF EXPERIMENTS

- 1) Installation of J2SDK
- 2) Write a program to show Scope of Variables
- 3) Write a program to show Concept of CLASS in JAVA
- 4) Write a program to show Type Casting in JAVA
- 5) Write a program to show How Exception Handling is in JAVA
- 6) Write a Program to show Inheritance
- 7) Write a program to show Polymorphism
- 8) Write a program to show Access Specifiers (Public, Private, Protected) in JAVA
- 9) Write a program to show use and Advantages of CONSTRUCTOR
- 10) Write a program to show Interfacing between two classes
- 11) Write a program to Add a Class to a Package
- 12) Write a program to show Life Cycle of a Thread
- 13) Write a program to demonstrate AWT.
- 14) Write a program to Hide a Class
- 15) Write a Program to show Data Base Connectivity Using JAVA
- 16) Write a Program to show "HELLO JAVA" in Explorer using Applet
- 17) Write a Program to show Connectivity using JDBC
- 18) Write a program to demonstrate multithreading using Java.
- 19) Write a program to demonstrate applet lifecycle.
- 20) Write a program to demonstrate concept of servlet.

REFERENCES

1. E. Balaguruswamy, "Programming In Java"; TMH Publications
2. The Complete Reference: Herbert Schildt, TMH
3. Deitel & Deitel, "JAVA, How to Program"; PHI, Pearson.
4. Cay Horstmann, Big JAVA, Wiley India.
5. Merlin Hughes, et al; Java Network Programming, Manning Publications/Prentice Hall

